A Counterexample to the Infinite-Dimensional Extreme-Geodesic Converse

Run fa_banach_001, agent_lane_04

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Status

This is a `counterexample_likely_valid` packet for arXiv:2512.21464, Ho Yun and Yoav Zemel, *Gaussian Optimal Transport Beyond Brenier's Theorem*. It answers negatively the infinite-dimensional converse left open after Theorem 6.11 and before Proposition 6.12: an extreme point of $\mathcal{K}_t(A, B)$ need not be a McCann interpolant.

Source Question

The source paper studies centered Gaussian covariance operators in the Bures-Wasserstein metric. In finite dimensions, Theorem 4.8 says that for singular covariances A, B, the extreme points of the barycenter set $\mathcal{K}_t(A, B)$ are exactly the McCann interpolants. On PDF page 38, the infinite-dimensional discussion says that every McCann interpolant is an extreme point, but that the converse remains open. The relevant page is included below.

Proof of Theorem 6.11.

1. Same to the proof of Theorem 4.10.
2. Let $\mathbf{\Gamma}_t = \mathbf{G}_t\mathbf{G}_t^* \in \mathcal{K}_t(\mathbf{A},\mathbf{B})$, where $\mathbf{G}_t = (1-t)\mathbf{A}^{1/2} + t\mathbf{M}$ with $\mathbf{M} \in \mathscr{G}(\mathbf{B}),\ \mathbf{A}^{1/2}\mathbf{M} \succeq \mathbf{0}$. To show $\mathbf{A}/\mathbf{\Gamma}_t = \mathbf{0}$, it suffices to show

$$\mathcal{R}(\mathbf{A}^{1/2}) \cap \mathcal{N}(\mathbf{\Gamma}_t) = \mathcal{R}(\mathbf{A}^{1/2}) \cap \mathcal{N}(\mathbf{G}_t^*) = \{\mathbf{0}\},$$

due to Lemma 2.4. Let $\mathbf{c} \in \mathcal{R}(\mathbf{A}^{1/2}) \cap \mathcal{N}(\mathbf{G}_t^*)$. Then, there exists some $\mathbf{d} \in \mathcal{H}$ such that $\mathbf{c} = \mathbf{A}^{1/2}\mathbf{d}$ and

$$0 = \mathbf{d}^*\mathbf{G}_t^*\mathbf{c} = \mathbf{d}^*\mathbf{G}_t^*\mathbf{G}_1\mathbf{d} = (1-t)\mathbf{d}^*[\mathbf{A}^{1/2}\mathbf{M}]\mathbf{d} + t\mathbf{c}^*\mathbf{c} \geq t\|\mathbf{c}\|^2 \quad \Rightarrow \quad \mathbf{c} = \mathbf{0}.$$

Consequently, we get $\mathbf{A}/\mathbf{\Gamma}_t = \mathbf{0}$. Repeating this with $\mathbf{B}$, we get $\mathbf{B}/\mathbf{\Gamma}_t = \mathbf{0}$.
3. Let $\tilde{\mathbf{\Gamma}} : [0,1] \to \mathcal{B}_1^+(\mathcal{H})$ be a constant-speed geodesic connecting $\mathbf{A}$ and $\mathbf{B}$. Fix $t \in (0,1)$. By the Cauchy-Schwarz inequality, we get

$$(1-t)\mathcal{W}_2^2(\mathbf{A},\mathbf{C}) + t\mathcal{W}_2^2(\mathbf{B},\mathbf{C}) \geq \left(\frac{1}{1-t} + \frac{1}{t}\right)^{-1} (\mathcal{W}_2(\mathbf{A},\mathbf{C}) + \mathcal{W}_2(\mathbf{B},\mathbf{C}))^2 \geq t(1-t)\mathcal{W}_2^2(\mathbf{A},\mathbf{B}).$$

From the proof of Proposition 6.8, the equality holds if and only if $\mathbf{C} \in \mathcal{K}_t(\mathbf{A},\mathbf{B})$. This shows that $\tilde{\mathbf{\Gamma}}_t \in \mathcal{K}_t(\mathbf{A},\mathbf{B})$, i.e., $\tilde{\mathbf{\Gamma}}_t = \mathbf{\Gamma}_t^{\mathbf{N}_{12}}$ for some $\mathbf{N}_{12}$ given in Theorem 6.9. Moving forward, from part 2 and Corollary 6.10, $\mathbf{\Gamma}^{\mathbf{N}_{12}} : [t,0] \to \mathcal{B}_1^+(\mathcal{H})$ (reversed time) is the unique constant-speed geodesic connecting $\mathbf{\Gamma}_t^{\mathbf{N}_{12}}$ and $\mathbf{A}$. Similarly, $\mathbf{\Gamma}^{\mathbf{N}_{12}} : [t,1] \to \mathcal{B}_1^+(\mathcal{H})$ is the unique constant-speed geodesic connecting $\mathbf{\Gamma}_t^{\mathbf{N}_{12}}$ and $\mathbf{B}$.

□

Analogous to Villani's quote [51] *"a geodesic in the space of laws is the law of a geodesic,"* our findings for Gaussian measures can be summarized as: *"A geodesic in the space of Green operators is the Green operator of a geodesic"*. Finally, we establish an infinite-dimensional counterpart to Theorem 4.8. While we show that every McCann interpolant is an extreme point of the set of barycenters, the converse remains an open question, especially due to (Case 2) in the proof of Theorem 4.8.

Proposition 6.12 (Extreme Geodesics)**.** *Let $\mathbf{A},\mathbf{B} \in \mathcal{B}_1^+(\mathcal{H})$ be covariance operators with $\mathbf{A} \rightsquigarrow \mathbf{B}$. The set of barycenters $\mathcal{K}_t(\mathbf{A},\mathbf{B})$, which is a closed convex set, has the following properties:*

1. *If $\mathbf{\Gamma}_t$ is a McCann interpolant at time t, then it is an extreme point of $\mathcal{K}_t(\mathbf{A},\mathbf{B})$.*
2. *For any barycenter $\mathbf{\Gamma}_t^{\mathbf{N}_{12}} \in \mathcal{K}_t(\mathbf{A},\mathbf{B})$, we have $\mathbf{\Gamma}_t^{\mathbf{N}_{12}}/\mathbf{A} = t^2(\mathbf{B}/\mathbf{A} - \mathbf{N}_{12}^*\mathbf{N}_{12})$.*
3. *$\mathbf{\Gamma}_t$ is a McCann interpolant at time t if and only if $\mathbf{\Gamma}_t/\mathbf{A} = \mathbf{0}$.*

Proof of Proposition 6.12.

1. Repeating the proof of Theorem 4.8,

$$\mathcal{A} := \left\{\mathbf{N}_{12} : \mathcal{H}_2 \to \mathcal{H}_1 \mid \mathcal{R}(\mathbf{N}_{12}) \subset \mathcal{N}(\mathbf{B}_{11}^{1/2}\mathbf{A}_{11}^{1/2}),\ \mathbf{N}_{12}^*\mathbf{N}_{12} \preceq \mathbf{B}/\mathbf{A}\right\},$$

is a closed convex set. Then, any $\mathbf{N}_{12} : \mathcal{H}_2 \to \mathcal{H}_1$ with $\mathcal{R}(\mathbf{N}_{12}) \subset \mathcal{N}(\mathbf{B}_{11}^{1/2}\mathbf{A}_{11}^{1/2}),\ \mathbf{N}_{12}^*\mathbf{N}_{12} = \mathbf{B}/\mathbf{A}$ is an extreme point of $\mathcal{A}$ since the equality in (24) holds if and only if $\mathbf{P}_0 = \mathbf{P}_1 \in \mathcal{E}$. Then, from Remark 4.2, it follows that $\mathcal{K}_t(\mathbf{A},\mathbf{B})$ is the closed convex set. It also follows that $\mathbf{\Gamma}_t \in \mathcal{K}_t(\mathbf{A},\mathbf{B})$ is an extreme point if and only if $\mathbf{N}_{12} \in \mathcal{A}$ is an extreme point, hence a McCann interpolant at time t is an extreme point due to Theorem 6.9.
2. Recall $\mathbf{T}_{11} = \mathbf{A}_{11}^{\dagger/2}(\mathbf{A}_{11}^{1/2}\mathbf{B}_{11}\mathbf{A}_{11}^{1/2})^{1/2}\mathbf{A}_{11}^{\dagger/2} : \mathcal{R}(\mathbf{A}_{11}^{1/2}) \subset \mathcal{H}_1 \to \mathcal{H}_1$ from Theorem 6.5, which is an s.p.d. operator due to Proposition 6.3. Using the block decomposition as in the proof of Theorem 6.9, we obtain

$$(\mathbf{\Gamma}_t)_{11} = \mathbf{X}_{11}\mathbf{X}_{11}^*, \quad (\mathbf{\Gamma}_t)_{12} = \mathbf{X}_{11}[t\mathbf{M}_{21}]^*, \quad (\mathbf{\Gamma}_t)_{22} = t^2\mathbf{B}_{22} = t^2[\mathbf{M}_{22}\mathbf{M}_{22}^* + \mathbf{M}_{21}\mathbf{M}_{21}^*],$$

where

$$\mathbf{X}_{11} := [(1-t)\mathbf{A}_{11}^{1/2} + t\mathbf{M}_{11}] = [(1-t)\mathbf{I}_{11} + t\mathbf{T}_{11}]\mathbf{A}_{11}^{1/2}.$$

Proof Intuition

The source parametrizes the infinite-dimensional barycenter set by operators

$$N : H_2 \to H_1, \qquad \mathcal{R}(N) \subseteq \mathcal{N}(B_{11}^{1/2} A_{11}^{1/2}), \qquad N^*N \le B/A.$$

McCann interpolants correspond to the boundary condition $N^*N = B/A$. After factoring $N = W(B/A)^{1/2}$, this is the unit ball of operators from the support of B/A into the defect space $\mathcal{N}(B_{11}^{1/2} A_{11}^{1/2})$. In infinite dimensions, the extreme contractions include coisometries as well as isometries. A non-isometric coisometry gives an extreme parameter N, but it has $N^*N < B/A$, so the corresponding barycenter is extreme and non-McCann.

Theorem

Counterexample Theorem. *There exist trace-class covariance operators $A, B \in \mathcal{B}_1^+(\mathcal{H})$ on a separable infinite-dimensional Hilbert space and $t \in (0,1)$ such that $A \rightsquigarrow B$ in the sense of Yun–Zemel, and $\mathcal{K}_t(A,B)$ has an extreme point which is not a McCann interpolant.*

Proof. Let H_1 and H_2 be two copies of $\ell_2(\mathbb{N})$, with orthonormal bases (e_n) and (f_n), and put $\mathcal{H} = H_1 \oplus H_2$. Define positive compact operators on the two summands by

$$A_{11}^{1/2} e_n = 2^{-n} e_n, \qquad C^{1/2} f_n = 2^{-n} f_n,$$

and define $B_{11}^{1/2}$ on H_1 by

$$B_{11}^{1/2} e_{2n-1} = 0, \qquad B_{11}^{1/2} e_{2n} = 2^{-n} e_{2n} \quad (n \ge 1).$$

Set

$$A = \begin{pmatrix} A_{11} & 0 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} B_{11} & 0 \\ 0 & C \end{pmatrix}.$$

Both are positive trace-class operators. Since $B_{12} = 0$, the A-Schur complement of B is $B/A = C$. Also

$$K := \mathcal{N}(B_{11}^{1/2} A_{11}^{1/2}) = \overline{\operatorname{span}}\{e_{2n-1} : n \ge 1\}.$$

The support $\overline{\mathcal{R}(C^{1/2})} = H_2$ and K are both separable infinite-dimensional Hilbert spaces, so there is an isometry from H_2 into K. Thus the dimensional condition in Theorem 6.5 of the source holds, and $A \rightsquigarrow B$.

Let P_o be the orthogonal projection of H_2 onto $\overline{\operatorname{span}}\{f_{2n-1} : n \ge 1\}$. Define

$$W : H_2 \to K, \qquad W f_{2n-1} = e_{2n-1}, \qquad W f_{2n} = 0.$$

Then $WW^* = I_K$, while $W^*W = P_o \ne I_{H_2}$. Thus W is a coisometry but not an isometry. Put

$$N := WC^{1/2} : H_2 \to H_1.$$

Then $\mathcal{R}(N) \subseteq K$ and

$$N^*N = C^{1/2} P_o C^{1/2} \le C = B/A,$$

with strict inequality because $C^{1/2} f_2 \ne 0$ but $P_o f_2 = 0$. Therefore N is an admissible parameter in the source's set

$$\mathcal{A} = \{R : H_2 \to H_1 : \mathcal{R}(R) \subseteq K,\ R^*R \le C\}.$$

We next prove that N is extreme in $\mathcal{A}$. First observe that every $R \in \mathcal{A}$ factors uniquely as $R = VC^{1/2}$, where $V : H_2 \to K$ is a contraction. Indeed, $R^*R \leq C$ gives $\|Rx\| \leq \|C^{1/2}x\|$, so $V(C^{1/2}x) = Rx$ defines a contraction on the dense range of $C^{1/2}$, and hence extends uniquely to H_2.

It remains to note that W is an extreme contraction. Write $H_2 = E \oplus F$, where $E = P_o H_2$ and $F = (I - P_o)H_2$. Suppose $W = (V_0 + V_1)/2$ with $V_0, V_1 : H_2 \to K$ contractions. For $x \in E$,

$$Wx = \frac{V_0 x + V_1 x}{2}, \qquad \|Wx\| = \|x\|,$$

and the strict convexity of the Hilbert norm implies $V_0 x = V_1 x = Wx$. If some $z \in F$ had $u := V_i z \neq 0$, choose $x = sW^*u \in E$ with $s > 0$. Since $V_i x = Wx = su$, contractivity would give

$$(s+1)^2\|u\|^2 = \|V_i(x+z)\|^2 \leq \|x+z\|^2 = s^2\|u\|^2 + \|z\|^2$$

for every $s > 0$, impossible for large s. Hence $V_i z = 0$ for all $z \in F$, and $V_i = W$. Thus W, and therefore $N = WC^{1/2}$, is extreme.

Choose

$$M_{22} := (C - N^*N)^{1/2}.$$

The parametrization in Theorem 6.9 of the source gives a barycenter $\Gamma_t = \Gamma_t^N \in \mathcal{K}_t(A, B)$. In the present diagonal example this can be written explicitly as $\Gamma_t = G_t G_t^*$, where

$$G_t = \begin{pmatrix} X_t & 0 \\ tN^* & tM_{22} \end{pmatrix}, \qquad X_t = (1-t)A_{11}^{1/2} + tB_{11}^{1/2}.$$

The operator X_t is injective for $0 < t < 1$. The map $N \mapsto \Gamma_t^N$ is affine and injective in this model, since its off-diagonal block is tX_tN. Hence extremality of N in $\mathcal{A}$ implies extremality of Γ_t^N in $\mathcal{K}_t(A, B)$.

Finally, Proposition 6.12 of the source gives

$$\Gamma_t^N / A = t^2(C - N^*N).$$

This Schur complement is nonzero. By the same proposition, a barycenter is a McCann interpolant if and only if its A-Schur complement vanishes. Thus Γ_t^N is an extreme point of $\mathcal{K}_t(A, B)$, but it is not a McCann interpolant. $\square$

Verification Notes

The construction uses no numerical computation. The crucial checks are:

- A and B are positive trace-class covariance operators.
- $A \rightsquigarrow B$ because the source's reachability defect space K and $\overline{\mathcal{R}((B/A)^{1/2})}$ are both separable infinite-dimensional.
- The chosen W is proved directly to be an extreme contraction.
- $N^*N < C = B/A$, so the source's own Schur-complement criterion rules out McCann interpolation.

Limitations and Novelty Check

This counterexample does not affect the finite-dimensional Theorem 4.8. It uses precisely the infinite-dimensional Case 2 obstruction identified by the source paper.

Cheap run indexes were searched for `2512.21464`, the source title, `Gaussian optimal transport`, `Brenier`, `Bures-Wasserstein`, `McCann interpolant`, and `extreme geodesics`. No exact prior packet was found. Bounded web search for the exact arXiv id, the title, `McCann`, and `extreme` surfaced the source paper and mirrors only, not a later resolution.

Human Review Recommendation

Reviewers should focus on two points: first, the transfer of extremality from the coisometry W to $N = WC^{1/2}$ through the dense range of $C^{1/2}$; second, the affine-injective parametrization $N \mapsto \Gamma_t^N$ in the source's Theorem 6.9/Proposition 6.12 framework.

References

[1] H. Yun and Y. Zemel, *Gaussian Optimal Transport Beyond Brenier's Theorem*, arXiv:2512.21464, 2025.